

The Keynesian Case for Trade Protection: A Modern Treatment*

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Abstract

We revisit the Keynesian rationale for tariffs: when downward nominal wage rigidity creates inefficient employment losses, tariffs can serve as second-best instruments by expanding labor demand in affected sectors. We characterize optimal tariffs in a dynamic multisector economy with wage rigidities and a flexible exchange rate. Our contribution is twofold. First, from an analytical standpoint, tariffs can be counterproductive instruments: they may reduce labor demand in the very sectors they are meant to protect, as currency appreciation depresses export demand and discourages current spending. As a result, optimal tariffs may be subsidies. Second, from a quantitative standpoint, optimal interventions and the associated welfare gains remain modest in our U.S. application even when labor distortions are sizable. Optimal tariffs are small because of their limited effectiveness in shifting employment across sectors relative to the distortions they create.

Keywords: Trade policy; tariffs; downward nominal wage rigidity; unemployment; exchange rates.

JEL classification: F13, F41, E24.

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When a free trader argues that a tariff cannot increase employment but can only divert employment from one industry to another, he is tacitly assuming that a man who loses his employment in one direction will lower the wage rate which he is willing to accept until he finds employment in another direction. When small changes only are in question, there may be much long-run truth in this. But in present circumstances it is sheer nonsense.

— John Maynard Keynes¹

1 Introduction

Tariffs are back on the policy menu, and politicians often justify them as a way to create jobs. But does expanding employment in certain sectors through tariffs improve aggregate welfare? Classical free-traders argue that such policies (beyond what is justified on terms-of-trade grounds) are distortionary: expanding employment in protected sectors draws workers away from other productive uses. However, Keynes challenged this view during the Great Depression: if downward nominal wage rigidity (DNWR) leaves employment inefficiently low and domestic subsidies are infeasible, there is a rationale for using tariffs to stimulate labor demand by shifting expenditure from imports to domestic goods. This paper revisits the Keynesian rationale for tariffs aimed at addressing DNWR-induced employment losses across sectors, asking whether such tariffs necessarily improve welfare and, when they do, how large the optimal tariffs and associated welfare gains are.²

Our analysis highlights two important caveats to the Keynesian case for tariffs under a flexible exchange rate, even under assumptions favorable to that case: no other policy instruments are available, and the government has full information. First, from an analytical standpoint, we find that a broad tariff intervention can backfire through the exchange-rate response, reducing employment in the very sectors it is intended to protect; optimal policy may instead subsidize imports. Second, from a quantitative standpoint, even when taxing imports is optimal, the magnitude of optimal

1. Keynes (1981, 490).

2. Microeconomic evidence documents substantial downward rigidity in nominal wages for both incumbent workers and new hires (Grigsby et al. 2021; Hazell and Taska 2025); see also Schaefer and Singleton (2023) and Jo (2025).

tariffs and their welfare gains remain modest in our U.S. application, across a variety of shocks that induce sizable labor wedges. We show that input–output linkages make even targeted tariffs blunt instruments for stimulating labor demand.

We develop a dynamic, multisector open-economy model with DNWR and characterize optimal trade policy under a flexible exchange rate. Labor is imperfectly substitutable across sectors, and its reallocation may be subject to mobility frictions. In the baseline, the central bank sets a path for consumer price inflation, which the government takes as given when choosing optimal tariffs; an extension with costly inflation allows monetary policy to respond endogenously to trade policy.³

We derive two analytical results. Our first analytical result is a Pigouvian formula for optimal trade policy in the spirit of Costinot and Werning (2026), applied to labor-market distortions. The Keynesian rationale for intervention rests on a sectoral *labor wedge*: when DNWR binds and employment is constrained by firms’ labor demand, the marginal value product of labor in the sector exceeds the marginal opportunity cost of supplying labor to the sector. The formula links the optimal tariff in each sector to how restricting imports in the sector affects employment in sectors with high labor wedges. This formula provides the organizing framework for the results that follow.

Our second analytical result shows that the optimal policy need not be an import tariff and characterizes when import subsidies are instead optimal. With a flexible exchange rate, tariffs affect labor demand through three channels. The first captures the standard Keynesian intuition familiar to policymakers: by raising import prices, tariffs induce import substitution toward home goods. However, the home currency appreciates in response to tariffs, offsetting the inflationary pressure from higher import prices by lowering the home-currency prices of tradable goods.⁴ The appreciation produces two contractionary forces in sectors where DNWR binds. First, since wages cannot adjust downward, the appreciation makes exports from these sectors more expensive abroad, reducing foreign demand. Second, through interest parity, it raises the domestic interest

3. In the extension, the central bank optimally balances inflation costs against employment gains, giving rise to a “dual mandate,” while the government chooses tariffs anticipating this monetary-policy response. Our analytical results continue to hold in this extension.

4. In a broad class of models, tariffs induce a real appreciation by raising the price of the domestic consumption bundle relative to world prices, measured in the same unit of account. Under CPI inflation targeting, this real appreciation occurs through a nominal appreciation rather than higher domestic inflation.

rate, inducing households to postpone consumption and reducing aggregate demand. When tariff intervention is narrowly targeted, the exchange-rate response is limited and the import-substitution channel dominates. By contrast, when the targeted sectors account for a sufficiently large share of the tradable economy, the two contractionary channels dominate, and tariffs reduce employment in the sectors where DNWR binds.⁵ Thus, when DNWR binds broadly across sectors, the Pigouvian formula implies that import subsidies, rather than tariffs, are optimal.

We then turn to the quantitative analysis to explore the magnitudes of Keynesian tariffs. We develop and calibrate a quantitative model to the 2024 U.S. economy using BEA input–output accounts and conduct two contrasting exercises.

In the first exercise, we consider a large demand-driven recession, modelled as a transitory discount-rate shock, that makes DNWR bind broadly and especially in nontraded sectors. As the theory predicts, the optimal policy subsidizes imports across sectors. We isolate the exchange-rate channel by showing that the sign of the optimal policy implied by the Pigouvian formula flips when we shut down the currency response. Nevertheless, import subsidies remain a poor substitute for first-best labor subsidies: the welfare gain from the intervention amounts to only 21 percent of the deadweight loss induced by DNWR.

In the second exercise, we instead consider a trade shock to a single industry. In particular, motivated by the rapid expansion of Chinese electric-vehicle production and increasing competition in global automobile markets, we subject the economy to a decline in world automotive prices. The shock generates a 70.9 percent labor wedge and leaves automotive employment 14.7 percent below its first-best level. Even so, the optimal automotive tariff is only 3.8 percent; the resulting welfare gain is only about 12 percent of the deadweight loss induced by DNWR. Because the automobile sector is small relative to the economy, the tariff-induced currency appreciation is negligible in this exercise. Instead, the limited optimal intervention reflects the weak transmission of import substitution into automotive employment. We find that input-output linkages are important in

5. In the one-sector limit, generalized Lerner symmetry (Lerner 1936; Costinot and Werning 2019) provides a complementary interpretation: a current-period import tariff is equivalent to a current-period export tax—which reduces export demand—combined with a tightening of capital controls—which depresses current consumption.

generating weak transmission. In theory, this could happen through two forces; demand generated by tariffs "leaks" to other activities via demand for inputs, while tariffs raise input prices and thus reduce sectoral demand. Our quantitative decomposition of these forces shows that the input-cost channel plays a modest quantitative role, while demand leakages through imported inputs and upstream sectors are central. When we shut down both leakages in the partial-equilibrium benchmark, the implied automotive tariff rises from 4.2 to 13.4 percent.

Overall, even under conditions favorable to their use, Keynesian tariffs offer limited welfare gains: they induce little employment reallocation relative to the distortions they create, and the exchange-rate response can make import subsidies preferable.

Related Literature The earlier trade literature studied commercial policy in economies with wage rigidity. Haberler (1950) showed that factor-price rigidity can cause trade to generate unemployment and reduce welfare, creating a potential case for protection. Johnson (1965) generalized this insight, showing that domestic instruments are first best, while tariffs are second best. Subsequent work characterized commercial policy in economies with minimum wages and unemployment (Brecher 1974a, 1974b; Dixit 1978; Neary 1980); much of this literature was motivated by persistent labor-market distortions in developing economies. We provide a modern treatment of this literature and derive a Pigouvian formula for optimal tariffs, which we use to characterize how the exchange-rate response and production networks shape their sign and magnitude.

A parallel Old Keynesian literature asked whether the Keynesian case for protection survived under flexible exchange rates. Beginning with Mundell (1961) and developed further by Chan (1978), Eichengreen (1981), and Krugman (1982), this literature argued that the exchange-rate response could reverse the expansionary effects of tariffs. Their reduced-form general-equilibrium closures, however, precluded welfare analysis. Our microfounded framework instead embeds the intertemporal approach to the current account (Obstfeld and Rogoff 1995), disciplining exchange-rate and trade-balance responses and allowing us to characterize optimal policy.

The modern literature combining trade policy and nominal rigidities remains comparatively sparse. Dávila et al. (2025) develop a general dynamic theory of optimal tariffs in heterogeneous-

agent economies and apply it in a HANK model with nominal rigidities. Their analysis emphasizes intertemporal terms-of-trade manipulation, risk sharing, and redistribution, while ours centers on how trade policy interacts with DNWR-induced labor-market distortions across sectors. Kim et al. (2026) study how China’s dollar peg amplified the U.S. labor-market effects of the China shock. As part of their analysis, they evaluate temporary uniform tariffs on Chinese goods under the peg. We instead characterize optimal sector-specific tariffs under a flexible exchange rate.⁶

A broader quantitative trade and spatial literature studies the role of DNWR in adjustment to trade and structural shocks. Rodríguez-Clare et al. (2025) show that DNWR can account for the large and persistent regional disparities in employment losses that Autor et al. (2013) document following the China shock. Subsequent studies of global supply-chain disruptions and the recent trade war likewise find material labor-market effects from DNWR (Ulate et al. 2025; Rodríguez-Clare et al. 2026). Whereas this literature focuses on the positive implications of DNWR for adjustment to trade and structural shocks, we study its normative implications for trade policy.⁷

Modern New Keynesian work studies the macroeconomic effects of tariffs and their interaction with monetary policy.⁸ Related to our contractionary mechanism, Auclert et al. (2025) derive conditions under which temporary tariff shocks are contractionary, emphasizing how they weaken export competitiveness and raise the current cost of goods. Whereas their analysis focuses on the aggregate output-gap and terms-of-trade consequences of tariff shocks, we study whether tariffs should be used to address preexisting sectoral labor wedges created by DNWR. A separate strand of the literature characterizes the optimal monetary-policy response to exogenous tariff shocks (Bianchi and Coulibaly 2025; Monacelli 2025; Werning et al. 2025). We instead take monetary policy as given and characterize the optimal trade-policy response to shocks that generate labor wedges through DNWR. Our multisector production-network structure further allows us to compare narrowly targeted with broad trade-policy interventions and to characterize how input–

6. Jeanne (2021) studies trade and currency wars under DNWR and low global demand, showing that, at the zero lower bound, import-tariff wars can reduce employment and generate self-fulfilling global liquidity traps.

7. See also O’Connor (2025), who study optimal place-based transfers in a spatial model with DNWR.

8. Auray et al. (2025) study how monetary regimes shape noncooperative tariff games.

output linkages shape the effects of tariffs on labor demand across sectors.⁹

Recent time-series evidence supports the exchange-rate mechanism in our model. Drawing on a narrative identification of U.S. tariff shocks, den Besten et al. (2026) document contractionary effects of tariffs. Most relevant for our setting, their postwar estimates—covering the transition toward increasingly flexible exchange rates—show an immediate real appreciation, a sharp decline in exports on impact, and more front-loaded contractions in output and manufacturing activity. Many other studies, using a range of samples and identification strategies, likewise find that tariff increases or tariff news lead to an appreciation of the country’s currency (Furceri et al. 2022; Barattieri et al. 2021; Jeanne and Son 2024; Matveev and Ruge-Murcia 2024; Ostry et al. 2025).¹⁰

Cross-sectional evidence from the 2018–2019 U.S. trade war is also consistent with our quantitative findings: employment gains from import protection were modest or absent, while tariffs on imported inputs reduced manufacturing employment and hiring demand (Flaaen and Pierce 2026; Javorcik et al. 2025).¹¹

Organization. Section 2 presents the model; Section 3 characterizes optimal trade policy; Section 4 presents the quantitative analysis; and Section 5 concludes.

9. Our analysis of policy responses to asymmetric sectoral shocks relates to Guerrieri et al. (2021, 2022), who focus on monetary and fiscal policy.

10. The dollar depreciation following the April 2025 “Liberation Day” announcement does not necessarily contradict this evidence. Anticipated retaliation can reverse the exchange-rate response to unilateral tariffs (Ostry et al. 2025); heightened policy uncertainty may also have eroded the safety premium on dollar assets, consistent with the concurrent rise in long-term Treasury yields (Jiang et al. 2025; Itskhoki and Mukhin 2025).

11. Autor et al. (2024) likewise find no employment gains from import tariffs across a broader set of sectors.

2 Model

We consider a dynamic small-open-economy model with multiple sectors. Within each sector, a domestic good is produced using labor and is imperfectly substitutable with the corresponding imported good. We introduce input–output linkages in an extension and in the quantitative analysis. Labor supplied to different sectors is imperfectly substitutable, reflecting workers’ heterogeneous comparative advantage and sectoral mobility frictions. The only source of distortion under *laissez-faire* is DNWR, which makes employment inefficiently low in the sectors where it binds. The central bank sets the path of inflation, while the government chooses sector-specific import tariffs.

2.1 Environment

The model is set in discrete time over an infinite horizon, with periods indexed by $t \in \mathcal{T} \equiv \{0, 1, \dots\}$. Within each period, we index sectors with $j \in \mathcal{J}$. For any variable z_{jt} , we let $\mathbf{z}_t = (z_{jt})_{j \in \mathcal{J}}$ denote the vector across all sectors, and $\mathbf{z}$ denote the sequence $\{\mathbf{z}_t : t \in \mathcal{T}\}$.

Household The economy features a representative household that consumes final goods and supplies labor across sectors. Its lifetime utility is

$$U_0 \equiv \sum_{t=0}^{\infty} \left(\prod_{s=0}^{t-1} \beta_s \right) u_t(C_t, L_t), \quad (1)$$

where C_t denotes aggregate consumption and L_t aggregate labor supply. The period utility function $u_t(\cdot, \cdot)$ is increasing in C_t , decreasing in L_t , and strictly concave in both arguments.¹²

Within each period, preferences have a nested structure: period consumption C_t depends on a vector of sectoral consumption indices $\mathbf{C}_t = (C_{jt})_{j \in \mathcal{J}}$, where each C_{jt} in turn depends on consumption of domestic and imported varieties within the sector. Formally,

$$C_t = H_t(\mathbf{C}_t), \quad C_{jt} = h_{jt}(d_{jt}, m_{jt}) \quad \text{for each } j \in \mathcal{J}, \quad (2)$$

where d_{jt} and m_{jt} denote household consumption of the domestic and imported varieties of good

12. Throughout, all functions characterizing preferences and technology are continuously differentiable.

j . The functions H_t and h_{jt} are increasing, strictly quasi-concave, and homogeneous of degree one in their consumption arguments.

The representative household also allocates aggregate labor supply L_t across sectors, with labor imperfectly substitutable between sectors. The labor supply feasibility set is given by

$$G_t(\ell_t; \ell_{t-1}) \leq L_t, \quad (3)$$

where $\ell_t = (\ell_{jt})_{j \in \mathcal{J}}$ denotes the vector of effective labor supplied to each sector in period t . The function G_t is jointly convex in both arguments.¹³ The curvature of G_t with respect to current allocations captures the degree of imperfect substitutability of labor across sectors, with greater curvature corresponding to lower substitutability. The linear case corresponds to perfect substitutability of labor across sectors, while the Leontief (max-function) limit corresponds to zero substitutability. Dependence on past allocations allows for sectoral mobility frictions. The formulation nests the static discrete-choice model of Roy (1951)—in which the curvature reflects heterogeneity in workers' comparative advantage across sectors—and dynamic discrete-choice models of the Artuç et al. (2010) type—in which dependence on past allocations arises from fixed costs of switching sectors.¹⁴

Downward Nominal Wage Rigidity The nominal wage w_{jt} in each sector j cannot fall below $\delta_j \in [0, 1]$ times its level in the previous period:

$$w_{jt} \geq \bar{w}_{jt} \equiv \delta_j w_{j,t-1}. \quad (4)$$

This formulation follows Schmitt-Grohé and Uribe (2016).

Domestic Technology Each sector j contains a unit mass of competitive firms that hire labor to produce output according to

$$y_{jt} \leq f_{jt}(\ell_{jt}), \quad (5)$$

13. We also assume that any feasible $\{\ell_t\}_{t \geq 0}$ is bounded whenever $\{L_t\}_{t \geq 0}$ is bounded and ℓ_{-1} is finite.

14. Appendix A.4 provides these constructions. The ACM-type construction assumes insurance against idiosyncratic switching costs so that it admits a representative-household formulation.

where ℓ_{jt} denotes labor employed in sector j and f_{jt} is an increasing, differentiable and concave function. Sector j 's output can be used for domestic consumption, or exported abroad:

$$y_{jt} = d_{jt} + x_{jt}, \quad (6)$$

where x_{jt} denotes exports in sector j . For our quantitative analysis, we extend this production structure to allow for standard multisector input–output production structures (Caliendo and Parro 2015).

International Trade Within each period, the country trades with the rest of the world at spot prices $\mathbf{p}_{xt}^*$ and $\mathbf{p}_{mt}^*$, expressed in foreign currency. Let ϵ_t denote the nominal exchange rate, measured in home currency per unit of foreign currency, so that an increase represents a depreciation.

We follow the small-open-economy (SOE) formulation in Galí and Monacelli (2005) and Demidova and Rodríguez-Clare (2013): in each sector, import supply is perfectly elastic at exogenous world prices p_{mjt}^* , while export demand is downward-sloping and isoelastic, with inverse demand

$$p_{xjt}^* = \rho_{xjt}^* x_{jt}^{-1/\sigma_j^*}, \quad (7)$$

where $\sigma_j^* > 1$ is the elasticity of export demand and ρ_{xjt}^* is the export demand shifter.¹⁵ Downward-sloping export demand allows currency appreciation to reduce exports smoothly, making the export-contraction channel central to the analysis in Section 3 amenable to first-order characterization.

Let p_{xjt} denote the export price in home currency. We assume that domestically owned export intermediaries charge a monopoly markup over the domestic producer price:

$$p_{xjt} \equiv \epsilon_t p_{xjt}^* = \frac{\sigma_j^*}{\sigma_j^* - 1} p_{djt}. \quad (8)$$

The intermediaries internalize the country's export market power, allowing us to abstract from the standard terms-of-trade motive for trade policy. Their profits accrue to the representative household.

15. The underlying idea behind these SOE models is that the country is an atomistic buyer in world markets but the sole producer of its own differentiated varieties, so it takes import prices as given while facing downward-sloping export demand. We could also allow for an upward-sloping, isoelastic import supply curve. Optimal policy would then include a sector-specific constant ad valorem tariff to manipulate the terms of trade; the optimal tariffs derived below should be interpreted as deviations from these baseline tariffs in response to DNWR-induced distortions.

Appendix A.1 provides the microfoundation for this pricing behavior.¹⁶

Bonds, Balance of Payments, and Interest Parity Across periods, the country can borrow and lend through one-period bonds denominated in foreign units of account, with exogenous gross interest rate R_t^* . Balance of payments requires the trade balance to equal net purchases of foreign assets:

$$p_{xt}^* \cdot x_t - p_{mt}^* \cdot m_t = \frac{B_t^*}{R_t^*} - B_{t-1}^*, \quad (9)$$

where B_t^* denotes net foreign assets held at the end of period t . The country begins with initial net foreign assets B_{-1}^* and faces the no-Ponzi condition $\liminf_{T \rightarrow \infty} \left(\prod_{s=0}^T (R_s^*)^{-1} \right) B_T^* \geq 0$.

The household may also trade a one-period nominal bond denominated in home currency and available in zero net supply. Let i_t denote the domestic nominal interest rate. No arbitrage between domestic and foreign bonds implies the interest parity condition

$$1 + i_t = \frac{\epsilon_{t+1}}{\epsilon_t} R_t^*. \quad (10)$$

Since the domestic bond is in zero net supply, we omit its holdings from the household problem.

Tariffs The government may levy a sector-specific tariff t_{jt} on imports.¹⁷ Let p_{mjt} denote the domestic-currency price of the sectoral imports. Under no-arbitrage,

$$p_{mjt} = \epsilon_t p_{mjt}^* + t_{jt}. \quad (11)$$

Both equations (8) and (11) assume full exchange-rate pass-through, which we relax in Section 3.4. All tariff revenue is rebated to the household through a lump-sum transfer T_t .¹⁸

16. The following construction ensures that each export intermediary behaves competitively in the home market but as a monopolist when exporting. The world market consists of a continuum of foreign destinations; a corresponding continuum of export intermediaries handles home exports, with each intermediary specializing in a distinct destination. Each intermediary is the exclusive supplier of the home variety in its destination and faces an isoelastic demand curve. Conversely, each intermediary is atomistic in the home market and takes the domestic price as given when sourcing goods. With a common elasticity across destinations, aggregation yields (7), while profit maximization yields (8).

17. Here, we express tariffs as specific (i.e., per-unit) taxes. This is without loss of generality, since any ad valorem tariff can be represented by the equivalent per-unit wedge.

18. Requiring budget balance every period is without loss of generality since Ricardian equivalence holds.

Inflation Targets Let P_t denote the Consumer Price Index (CPI), namely the price of the consumption aggregate C_t , and let $\pi_t \equiv \ln(P_t/P_{t-1})$ denote CPI inflation.¹⁹ The central bank sets a path for CPI inflation, $\pi \equiv (\pi_t)_{t \in \mathcal{T}}$, through monetary policy. This choice of price index follows central-bank practice, in which inflation objectives are typically defined in terms of consumption-based price indices.²⁰ Under standard assumptions, the desired path can be supported by an appropriate nominal interest-rate rule in the cashless limit of a monetary economy (Woodford 2003). Section 3.4 explores inflation targets based on alternative price indices.

2.2 Equilibrium

We now characterize the decentralized equilibrium with tariffs and DNWR.²¹ Apart from the export intermediaries' market power in foreign markets described above, all product and factor markets are competitive.

Firms' Problem In each sector j and period t , competitive firms choose labor inputs to maximize profits given wages and output prices:

$$\Pi_{jt} \equiv \max_{\ell_{jt} \geq 0} \{p_{djt} f_{jt}(\ell_{jt}) - w_{jt} \ell_{jt}\}. \quad (12)$$

Let ℓ_{jt}^d denote firms' labor demand. At an interior optimum, profit maximization implies

$$w_{jt} = p_{djt} f'_{jt}(\ell_{jt}) \quad \text{whenever } \ell_{jt} > 0, \quad (13)$$

Household's Problem Wages are determined in competitive labor markets. When DNWR binds in sector j , however, employment becomes demand-determined: the representative household

19. Formally, the CPI is defined by the household's expenditure-minimization problem within each period.

20. The Federal Reserve defines its inflation objective using the personal consumption expenditures price index, while the European Central Bank uses the Harmonised Index of Consumer Prices (Federal Open Market Committee 2025; European Central Bank 2025).

21. Throughout the theoretical analysis, we assume that the equilibria under consideration are interior with respect to consumption, production, and trade quantities.

cannot supply more labor than firms demand. Formally, the household solves²²

$$\begin{aligned}
V_t(B_{t-1}^*, \ell_{t-1}) &\equiv \max_{(C_t, L_t, B_t^*, \ell_t) \in \Gamma_t} u_t(C_t, L_t) + \beta_t V_{t+1}(B_t^*, \ell_t) \\
\text{subject to } & P_t C_t + \epsilon_t \frac{B_t^*}{R_t^*} \leq w_t \cdot \ell_t + \Pi_t + T_t + \epsilon_t B_{t-1}^*, \\
& G_t(\ell_t; \ell_{t-1}) \leq L_t, \\
& \ell_{jt} \leq \ell_{jt}^d \quad \text{if } w_{jt} = \bar{w}_{jt}, \quad \text{for each } j \in \mathcal{J}.
\end{aligned} \tag{15}$$

Here, Γ_t denotes the set of admissible choices, including a natural borrowing limit, and Π_t denotes total profits accruing to the household, including profits of domestic producers and export intermediaries. The first two constraints are standard: the household budget constraint and the labor supply feasibility constraint. The third is the additional constraint introduced by DNWR. When it binds, the wage is above its market-clearing level and employment is limited by firms' demand.

To characterize intertemporal consumption choice, let $R_t \equiv (1+i_t)(P_t/P_{t+1})$ denote the domestic real interest rate. Expressed in terms of real returns, the interest parity condition (10) becomes

$$R_t = \frac{P_t/\epsilon_t}{P_{t+1}/\epsilon_{t+1}} R_t^*. \tag{16}$$

The household's Euler equation is

$$\partial_C u_t = \beta_t R_t \partial_C u_{t+1}. \tag{17}$$

Because saving and borrowing take place in foreign currency at the world gross interest rate R_t^* , the relevant price for intertemporal choice is the CPI expressed in foreign currency, P_t/ϵ_t . We refer to its reciprocal, ϵ_t/P_t , as the real exchange rate.

Government Budget Tariff revenues are rebated lump-sum to the household, so the government budget satisfies

$$T_t = \sum_{j \in \mathcal{J}} t_{jt} m_{jt}. \tag{18}$$

22. The household also solves the expenditure-minimization problem

$$e_t(\mathbf{p}_{dt}, \mathbf{p}_{mt}, C_t) \equiv \min_{\mathbf{d}_t, \mathbf{m}_t \geq 0} \{ \mathbf{p}_{dt} \cdot \mathbf{d}_t + \mathbf{p}_{mt} \cdot \mathbf{m}_t \} \quad \text{subject to} \quad H_t((h_{jt}(d_{jt}, m_{jt}))_{j \in \mathcal{J}}) \geq C_t. \tag{14}$$

Definition 1 (Equilibrium). For initial conditions $(B_{-1}^*, \ell_{-1}, w_{-1}, P_{-1})$ and given paths of tariffs $\{t_t\}_{t \in \mathcal{T}}$ and inflation $\{\pi_t\}_{t \in \mathcal{T}}$, an equilibrium consists of sequences of prices and wages $\{(p_{dt}, p_{xt}, p_{mt}, p_{xt}^*, P_t, w_t)\}$, wage floors $\{\bar{w}_t\}$, nominal exchange rates $\{\epsilon_t\}$, nominal interest rates $\{i_t\}$, allocations $\{(C_t, L_t, C_t, \ell_t, y_t, x_t, m_t, d_t)\}$, foreign assets $\{B_t^*\}$, lump-sum transfers $\{T_t\}$, and profits $\{\Pi_t\}$, all indexed by $t \in \mathcal{T}$, such that: the household solves problem (15) and its expenditure-minimization problem (14); domestic firms solve the profit-maximization problem (12); export demand and export pricing satisfy (7) and (8); interest parity and import pricing satisfy (10) and (11); gross goods markets clear according to (6); the government budget satisfies (18); wage floors evolve according to (4); and the price level satisfies $\ln(P_t/P_{t-1}) = \pi_t$.

2.3 DNWR, Labor Wedges, and Misallocation

To characterize the distortion created by DNWR, consider first the household's labor-supply condition implied by problem (15) under flexible wages:

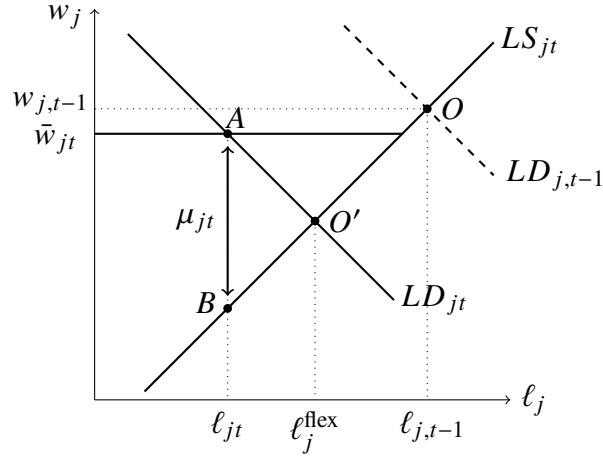
$$w_{jt} = \underbrace{\partial_{\ell_{jt}} G_t \left(\frac{-\partial_L u_t}{\partial_C u_t} P_t \right)}_{\text{current marginal disutility}} - \underbrace{\frac{1}{1+i_t} \left(-\partial_{\ell_{jt}} G_{t+1} \right) \left(\frac{-\partial_L u_{t+1}}{\partial_C u_{t+1}} P_{t+1} \right)}_{\text{continuation value}}. \quad (19)$$

Here, $\partial_{\ell_{jt}} G_t$ is the derivative with respect to current sectoral labor supply, whereas $\partial_{\ell_{jt}} G_{t+1}$ is the derivative with respect to lagged sectoral labor supply. Under flexible wages, the household chooses sectoral labor so that the wage equals the associated marginal opportunity cost, which is given by the current marginal disutility of supplying labor to the sector (expressed in monetary units) net of the continuation value arising from history dependence in sectoral employment movements.

When DNWR binds in sector j , the household is rationed: the marginal worker would like to supply more labor to the sector but cannot. The wage—and hence, by (13), the marginal value product of labor—therefore exceeds the marginal opportunity cost of supplying labor to that sector. We define the *labor wedge*, measured in monetary units, as the gap between the marginal value product of labor and its marginal opportunity cost:

$$\mu_{jt} \equiv p_{djt} f'_{jt}(\ell_{jt}) - \partial_{\ell_{jt}} G_t \left(\frac{-\partial_L u_t}{\partial_C u_t} P_t \right) + \frac{1}{1+i_t} \left(-\partial_{\ell_{jt}} G_{t+1} \right) \left(\frac{-\partial_L u_{t+1}}{\partial_C u_{t+1}} P_{t+1} \right). \quad (20)$$

Figure 1: Labor wedge under binding DNWR.



Notes: LD_{jt} is the marginal value product of labor given in (13), while LS_{jt} is the marginal opportunity cost of sectoral labor given by the right-hand side of (19). O denotes the initial equilibrium, O' the flexible-wage equilibrium after the decline in labor demand, and A the equilibrium under binding DNWR.

A positive labor wedge indicates that employment in sector j is inefficiently low; $\mu_{jt} = 0$ whenever DNWR does not bind.

Figure 1 illustrates the labor wedge. Here, we consider an inward shift in the labor-demand curve from $LD_{j,t-1}$ to LD_{jt} in sector j ; these curves are given by the marginal value product of labor in (13). The labor-supply curve LS_{jt} , given by the right-hand side of (19), represents the marginal opportunity cost of supplying labor to sector j . Under flexible wages, the shift in labor demand moves the equilibrium from O to O' , and employment falls to ℓ_j^{flex} . If DNWR instead constrains the wage at $\bar{w}_{jt}$, the equilibrium moves to A , with lower employment $\ell_{jt} < \ell_j^{\text{flex}}$. Point B gives the marginal opportunity cost at employment ℓ_{jt} , so the distance AB measures the labor wedge μ_{jt} . The triangle $O'AB$ represents the deadweight loss.

The household can respond to the decline in sectoral employment along two margins. First, it may reduce total employment L_t . Second, it may reallocate labor to other sectors where DNWR does not bind. Reallocation may leave total employment unchanged, but the resulting allocation of labor across sectors is inefficient. Intuitively, the DNWR forces workers to move to sectors where their skills are less suited. The greater the curvature of G_t , the less substitutable workers are across sectors and, therefore, the larger the efficiency loss from a given sectoral misallocation.²³

23. Our representative-household formulation abstracts from two other potential sources of inefficiency for tractability.

In Figure 1, lower substitutability would be represented by a steeper LS_{jt} curve, which produces a larger labor wedge for a given inward shift in labor demand.

In a constrained sector, employment is demand-determined and can increase only through higher demand for its output, $d_{jt} + x_{jt}$. Thus, the effect of any policy on the sector's employment operates through its impact on total demand from domestic and foreign buyers.

3 Optimal Trade Policy

In this section, we study tariffs as second-best instruments for addressing labor wedges induced by DNWR. In principle, sector-specific labor subsidies could eliminate these wedges by lowering firms' effective labor costs, while sufficiently high inflation could prevent the wage floors from binding in the first place.²⁴ We give the Keynesian rationale a favorable setting: labor subsidies are unavailable, and the government takes the inflation target set by the central bank as given. Even under these conditions, however, tariffs can be weak or counterproductive instruments for raising employment in sectors where DNWR binds.²⁵

Our analysis is not tied to a particular shock that causes DNWR to bind: it applies whenever market-clearing wages fall below the wage floors $\bar{w}_t$, whether due to a trade shock, a technology shock (e.g., automation), or a preference shock (e.g., a discount-rate shock).

3.1 Pigouvian Formula for Optimal Policy

We first derive the first-order necessary conditions for optimal trade policy. These conditions yield a Pigouvian formula for optimal tariffs, which encapsulates the Keynesian rationale for trade policy.

First, when individual workers sort across sectors based on expected earnings and the probability of rationing, they may fail to internalize how their choices affect the rationing faced by others, generating a congestion externality (Guerrieri et al. 2021). Second, our formulation assumes an efficient rationing technology. Viewed through the lens of a Roy model, workers with the weakest comparative advantage in a constrained sector are displaced first; if workers with stronger comparative advantage are also displaced, the distortion will be larger.

24. Given the path of inflation targets, the first-best labor subsidy in each sector equals the gap between the wage floor implied by DNWR and the corresponding wage under the flexible-wage equilibrium.

25. Appendix E contains the derivations and proofs for the results in this section.

Planner's Problem Let $\mathcal{F}_t \subset \mathcal{J}$ denote the sectors in which tariffs are available in period t , and let $\mathbf{t}_{\mathcal{F},t} \equiv (t_{jt})_{j \in \mathcal{F}_t}$ denote the corresponding tariff vector.²⁶ In the baseline, the planner chooses the path of tariffs $\mathbf{t}_{\mathcal{F}} \equiv (\mathbf{t}_{\mathcal{F},t})_{t \in \mathcal{T}}$ to maximize welfare (1) subject to the equilibrium conditions in Definition 1, taking the inflation targets set by the central bank as given.²⁷

Wedges and Variation in Welfare There are two potential sources of inefficiency in the economy: labor wedges, μ_{jt} , and import tariffs, t_{jt} . These wedges are expressed in terms of spot prices, but it will be convenient to express them in common period-0 prices. Specifically, we define

$$\mu_{jt}^0 \equiv \prod_{s=0}^{t-1} (1 + i_s)^{-1} \mu_{jt}, \quad t_{jt}^0 \equiv \prod_{s=0}^{t-1} (1 + i_s)^{-1} t_{jt}.$$

Now consider a policy variation. Its first-order welfare effect, in period-0 monetary units, is

$$d\mathcal{W}_0 = \underbrace{\sum_{t=0}^{\infty} \sum_{j \in \mathcal{J}} \mu_{jt}^0 d\ell_{jt}}_{\text{DNWR-induced distortions}} + \underbrace{\sum_{t=0}^{\infty} \sum_{j \in \mathcal{F}_t} t_{jt}^0 dm_{jt}}_{\text{tariff-induced distortions}}. \quad (21)$$

The first term captures the welfare effect of employment changes in sectors with binding DNWR. In these sectors, employment is demand-determined and inefficiently low, so $\mu_{jt}^0 > 0$. A policy that raises demand for the sector's domestic output therefore increases employment in that sector and improves welfare. The second term captures the marginal efficiency loss from tariff-induced distortions in consumption and production (a fiscal externality). With an existing positive tariff wedge, $t_{jt}^0 > 0$, reducing imports further exacerbates these distortions and creates a welfare loss.²⁸

Pigouvian Optimal Policy Formula We now characterize the optimal policy. Recall that $\boldsymbol{\ell} \equiv (\ell_t)_{t \in \mathcal{T}}$ denotes the path of sectoral employment. Let $\mathbf{m}_{\mathcal{F},t} \equiv (m_{jt})_{j \in \mathcal{F}_t}$ denote imports in sectors subject to tariffs in period t , and let $\mathbf{m}_{\mathcal{F}} \equiv (\mathbf{m}_{\mathcal{F},t})_{t \in \mathcal{T}}$ denote the corresponding path.

Optimal policy requires $d\mathcal{W}_0 = 0$ for any feasible variation in $(\boldsymbol{\ell}, \mathbf{m}_{\mathcal{F}})$. When DNWR does not bind and labor wedges are absent, setting all tariffs to zero satisfies this condition, implying

26. For example, tariffs may be unavailable in services sectors.

27. This does not preclude the central bank from adjusting its inflation targets in response to shocks. We only assume that the government takes these revised targets as given. The problem can therefore be interpreted as the government's best response to the central bank's choice of inflation targets in a Nash game between the two authorities.

28. Recall that export intermediaries internalize the country's market power, so no ToT externality appears in (21).

that free trade is optimal. This encapsulates the argument of the classical free-traders: tariffs are merely distortionary under flexible wages, absent terms-of-trade externalities.

When DNWR binds and labor wedges are present, however, the Keynesian rationale for tariffs arises: trade policy may improve welfare to the extent that it expands employment in sectors with labor wedges. Yet, since the planner cannot control employment and imports independently, it must weigh the welfare improvement from these employment gains against the distortion created by trade policy. In particular, the planner can control the path of imports in sectors subject to tariffs by adjusting these tariffs, but sectoral employment is determined through the equilibrium system.²⁹ We can therefore think of the planner as choosing $\mathbf{m}_{\mathcal{F}}$, with ℓ responding endogenously conditional on the central bank's inflation targets π .

Formally, we write the equilibrium employment path as $\ell(\mathbf{m}_{\mathcal{F}}; \pi)$ and consider the response

$$-\left. \frac{\partial \ell_{jt}}{\partial m_{is}} \right|_{d\pi=0}.$$

This expression captures how employment in sector j at time t responds to a restriction in imports in sector $i \in \mathcal{F}_s$ at time s , holding the inflation path fixed. This variation requires adjusting the entire path of tariffs to reduce m_{is} while leaving all other imports in $\mathbf{m}_{\mathcal{F}}$ unchanged. Assuming differentiability at the optimum, applying equation (21) to each import variation gives the first-order condition $\partial \mathcal{W}_0 / \partial m_{is} |_{d\pi=0} = 0$, which yields the Pigouvian policy formula below.³⁰

Theorem 1. *For each sector $i \in \mathcal{F}_s$ and period $s \in \mathcal{T}$, the optimal tariff satisfies*

$$t_{is}^0 = \sum_{t=0}^{\infty} \sum_{j \in \mathcal{J}} \mu_{jt}^0 \left(- \left. \frac{\partial \ell_{jt}}{\partial m_{is}} \right|_{d\pi=0} \right). \quad (22)$$

All derivatives are evaluated at the optimum.

The tariff formula (22) gives optimal trade policy a Pigouvian interpretation (Costinot and Werning 2023). Consider a sector $i \in \mathcal{F}_s$. If restricting imports in that sector raises employment

29. We assume that the mapping from tariffs to imports in sectors subject to trade policy is locally invertible.

30. DNWR can generate kinks in the equilibrium employment response where wage constraints become active or inactive. For simplicity, the theorem below assumes that $\ell(\mathbf{m}_{\mathcal{F}}; \pi)$ is differentiable at the optimum. More generally, at kinks, the optimality conditions must be formulated using directional derivatives.

in sectors with labor wedges across periods—that is, if the right-hand side of (22) is positive—the planner imposes a *Pigouvian import tax* on the sector. Conversely, if restricting imports reduces employment in sectors with labor wedges, the planner sets a *Pigouvian import subsidy*. We use this formula to sign optimal tariffs or subsidies in the next subsection.

3.2 Should We Tax or Subsidize Imports?

Theorem 1 might seem to provide a strong justification for Keynesian protectionism in response to DNWR-induced employment losses. However, it says little about the *sign* of the optimal tariff. We next show that optimal policy may entail import subsidies.

We first obtain a sharp closed-form characterization under some simplifying conditions to illustrate the key mechanisms; we relax these restrictions in later sections.

Definition 2 (As-If Two-Period Economy). An economy is *as-if two-period* if DNWR may bind only in period 0 and the following conditions hold from period 1 onward: preferences, technologies, and external conditions are constant; $\beta_t R_t^* = 1$; and G_t does not depend on past labor allocations.

These assumptions make the economy effectively a two-period model, with allocations in a steady state from period $t = 1$ onward. We refer to period $t = 0$ as the short run and periods $t \geq 1$ as the long run. Because the analysis focuses on the short run, we omit short-run time subscripts and use the subscript LR for long-run allocations.

Definition 3 (As-If Two-Sector Economy). An economy is *as-if two-sector* if its sectors can be partitioned into two nonempty sets, $\mathcal{A}$ and $\mathcal{B}$, such that sectors are identical within each set.

We consider an equilibrium in which DNWR binds in every type- A sector but in no type- B sector. One interpretation is that an adverse shock puts downward pressure on wages in type- A sectors but not in type- B sectors. The combined share of type- A sectors then captures how broadly the shock affects the economy: a small share represents a shock concentrated in a narrow segment of the economy, while a large share represents one affecting a broad range of sectors. Because sectors are identical within each set, the economy can be aggregated into two representative sector

types, A and B ; henceforth, variables with subscripts A and B denote the corresponding type-level aggregates. We relax the two-type restriction in Section 3.4.

To state the result, let $\Omega_j \equiv P_j C_j / (PC)$ denote the expenditure share of sector type j , $\omega_j \equiv p_{mj} m_j / (P_j C_j)$ its within-type import share, and $\psi_j \equiv x_j / y_j$ its export share. Let σ_j denote the within-sector elasticity of substitution between home and foreign varieties, Σ the elasticity of substitution across sector types, and γ the intertemporal elasticity of substitution (IES).³¹ We assume $\sigma_j > \Sigma$ throughout.³² Further, define

$$\check{\gamma} \equiv - \left. \frac{d \ln C}{d \ln(P/\epsilon)} \right|_{d\mathbf{m}_{LR}=0} \in (0, \gamma). \quad (23)$$

The elasticity $\check{\gamma}$ measures the decline in short-run consumption caused by a short-run real appreciation, taking the equilibrium response of long-run prices into account.³³ In line with the Pigouvian formula, the elasticity holds long-run imports fixed by varying the long-run tariffs accordingly. Finally, we assume that aggregate labor supply L is fixed for tractability.

Theorem 2 (Sign of the Optimal Import Taxes). *Consider an as-if two-period, two-sector economy with no intermediate inputs and fixed L . If, at the optimum, DNWR binds in type-A sectors but not in type-B sectors, then the optimal tariff on type-B imports is an import subsidy, $t_B < 0$, while the optimal tariff on type-A imports is an import subsidy if and only if $\Omega_A > \bar{\Omega}_A$, where the cutoff $\bar{\Omega}_A$ is defined by*

$$\frac{\bar{\Omega}_A}{1 - \bar{\Omega}_A} \equiv \frac{(1 - \psi_A)(\sigma_A - \Sigma)}{\psi_A \sigma_A^* + (1 - \psi_A)\check{\gamma}} \Xi_B, \quad (24)$$

where

$$\Xi_B \equiv \frac{(1 - \psi_B)\sigma_B\check{\gamma} + \psi_B\sigma_B^* [\sigma_B(1 - \omega_B) + \check{\gamma}\omega_B]}{(1 - \psi_B)\sigma_B\Sigma + \psi_B\sigma_B^* [\sigma_B(1 - \omega_B) + \Sigma\omega_B]} > 0. \quad (25)$$

All shares and elasticities are evaluated at the optimum.³⁴

31. These elasticities generally vary with the allocation unless preferences feature nested CES.

32. If $\sigma_A < \Sigma$, consumers substitute across sector types more readily than between home and foreign varieties within a sector. A higher p_{mA} then reduces d_A , so a positive tariff is never optimal.

33. By the real interest parity condition (16), P/ϵ and P_{LR}/ϵ_{LR} are the relevant prices for intertemporal consumption choice. A higher P/ϵ leads the household to substitute toward future consumption; the resulting increase in long-run aggregate demand raises P_{LR}/ϵ_{LR} . Hence, the real interest rate rises less than P/ϵ does, and $\check{\gamma} < \gamma$.

34. Appendix B.6 discusses optimal long-run tariffs.

Thus, when sectors with binding DNWR account for a sufficiently small share of expenditure, the optimal policy combines targeted tariffs in those sectors with subsidies elsewhere; when DNWR binds sufficiently broadly, subsidies are optimal in all sectors.

The logic follows from the Pigouvian formula (22), which in this setting reduces to

$$t_j = \mu_A \left(- \frac{\partial \ell_A}{\partial m_j} \bigg|_{d\pi=0} \right), \quad j \in \{A, B\}. \quad (26)$$

Since $\mu_A > 0$, the optimal tariff on type- j imports has the same sign as the response of type- A employment to a restriction on those imports, $-\partial \ell_A / \partial m_j|_{d\pi=0}$.

For type- A imports, three forces determine this employment response: the gain from import substitution and two contractionary effects arising from the exchange-rate response to tariffs. The import substitution channel is intuitive: a tariff-induced increase in the type- A import price p_{mA} shifts domestic demand toward home goods in the sector. The strength of this channel is governed by the domestic-sales share, $1 - \psi_A$, times the domestic-demand elasticity, $(\sigma_A - \Sigma)\omega_A$.

However, in general equilibrium, the tariff causes the currency to appreciate. To understand why, consider the inflation-targeting condition $d\pi = 0$. By Shephard's lemma,

$$0 = d\pi = \Omega_A [(1 - \omega_A)d \ln p_{dA} + \omega_A d \ln p_{mA}] + \Omega_B d \ln P_B. \quad (27)$$

A tariff-induced increase in p_{mA} puts upward pressure on the CPI. Since DNWR limits downward adjustment in type- A producer prices, maintaining $d\pi = 0$ requires P_B to fall. Currency appreciation delivers this decline by lowering the home-currency prices of type- B tradables through the pricing relations (8) and (11). The coefficient Ξ_B in (25) measures the pass-through from the exchange rate to P_B for a given labor allocation, taking equilibrium responses in foreign-currency prices into account.³⁵ Thus, the appreciation required to stabilize inflation is governed by $\Omega_A \omega_A / (\Omega_B \Xi_B)$, where the numerator measures the inflationary pressure from the higher p_{mA} and the denominator measures how effectively appreciation lowers the CPI.

The appreciation creates two contractionary forces on labor demand in type- A sectors. First,

35. When computing this pass-through, p_{xB}^* adjusts to clear the type- B labor market, while t_B adjusts to hold m_B fixed, as required by the derivative in the Pigouvian formula.

since wages cannot fall in these sectors, appreciation makes type- A goods more expensive abroad through (8), reducing export demand. The strength of this export-contraction channel is governed by the export share, ψ_A , times the export-demand elasticity, σ_A^* . Second, under CPI inflation targeting, appreciation raises the short-run domestic real interest rate through interest parity (16), further reducing domestic demand.³⁶ The strength of this intertemporal channel is governed by the domestic-sales share, $1 - \psi_A$, times the intertemporal consumption response, $\check{\gamma}$.

Putting these forces together, the planner compares the direct import-substitution effect, $(1 - \psi_A)(\sigma_A - \Sigma)\omega_A$, with the contractionary effects of appreciation, $[\psi_A\sigma_A^* + (1 - \psi_A)\check{\gamma}]\Omega_A\omega_A/(\Omega_B\Xi_B)$. The latter effects are stronger the larger Ω_A is. The comparison therefore yields the cutoff condition (24), after canceling the common factor ω_A .

The type- B subsidy operates through the same exchange-rate mechanism. An import subsidy in type B is deflationary and therefore induces depreciation when inflation is held fixed. This raises type- A exports and aggregate demand, increasing employment in the sectors with binding DNWR.

When a sectoral shock causes DNWR to bind in a small segment of the economy, tariffs targeted at the affected sectors create limited inflationary pressure and induce only a modest appreciation. The import-substitution effect then dominates, so the targeted tariffs raise employment in those sectors. By contrast, when a common shock causes DNWR to bind sufficiently broadly, tariffs imposed across the affected sectors generate substantial inflationary pressure and induce a large appreciation. The resulting contraction in exports and aggregate demand outweighs the employment gains from import substitution, so tariffs reduce employment in the very sectors they are intended to protect. The Pigouvian formula then implies that import subsidies are optimal across sectors.

3.3 Discussion

One-Sector Limit and Lerner Symmetry The tariff on type- A imports has a definite sign in the one-sector limit:

36. Put differently, a temporary tariff makes current consumption more expensive relative to future consumption—as reflected by the real appreciation (a fall in ϵ at fixed π)—inducing intertemporal substitution toward future consumption.

Corollary 3 (One Sector). *The optimal tariff on type-A imports is strictly negative in the limit where type-A sectors comprise the entire economy ($\Omega_B \rightarrow 0$).³⁷*

It may seem surprising at first that exchange-rate appreciation would be large enough to more than offset the import-substitution effect of an import tariff. Lerner symmetry helps clarify this result. The Lerner symmetry theorem states that uniform import tariffs (or subsidies) are equivalent to uniform export taxes (or subsidies) in equilibrium (Lerner 1936; Costinot and Werning 2019).

First, consider the limit $\gamma \rightarrow 0$ (which implies $\check{\gamma} \rightarrow 0$). In this limit, Lerner symmetry holds period by period, so a uniform import tariff in the short run is equivalent to a uniform export tax in the same period. As $\Omega_B \rightarrow 0$, the economy converges to a one-sector model, in which a tariff on type-A imports effectively becomes a uniform import tariff. It is therefore equivalent to an export tax, which reduces foreign demand for home output and depresses labor demand in type-A sectors. Conversely, an import subsidy is equivalent to an export subsidy and thereby serves as a proxy for the first-best labor subsidy. Hence, subsidizing imports is optimal in this limit, whereas imposing a tariff backfires.³⁸

With a nonzero IES, Lerner symmetry no longer holds period by period. However, the generalized result in Costinot and Werning (2019) implies that a short-run import tariff is equivalent to an export tax in the same period, combined with a capital control that discourages net inflows.³⁹ Such a capital control raises the domestic interest rate, discouraging short-run consumption. Intertemporal substitution therefore makes raising the short-run tariff even more undesirable in the one-sector limit. Intuitively, a temporary tariff makes current consumption more expensive relative to future consumption—as reflected by the real appreciation (a fall in ϵ at fixed π)—thereby further depressing short-run aggregate demand and employment in type-A sectors.

37. These and subsequent limits are obtained through simultaneous changes in the preferences $H(\cdot)$ and $h_j(\cdot)$, the production set $G(\cdot)$, and the export demand functions. We assume throughout that $\omega_A > 0$ and that the sectoral elasticities σ_j remain bounded.

38. When inflation π (and therefore P) is held constant, our model is isomorphic to one with a *real* (rather than nominal) wage rigidity that keeps the type-A real wage above the floor $\bar{w}_A/P$. In view of Costinot and Werning (2019), the Lerner symmetry theorem applies in this setting, treating any wedges that arise at the optimum as “taxes.”

39. Formally, Costinot and Werning (2019) show that uniform ad valorem import tariffs in period $t = 0$ are equivalent to uniform export taxes in periods $t \geq 0$ and uniform import subsidies of the same magnitude in periods $t \geq 1$. From period $t = 1$ onward, these export taxes and import subsidies are equivalent to a capital control that taxes inflows and subsidizes outflows between periods 0 and 1 (Costinot et al. 2014). Appendix B.1 formalizes this equivalence.

The exchange-rate movement highlighted above is precisely the general-equilibrium force underlying these Lerner symmetry results. The appreciation makes home goods more expensive abroad, just as an export tax would. Intertemporally, it raises the domestic interest rate and depresses current aggregate demand, just as a tighter capital control would.

Additional Limit Cases We explore two more limit cases that provide further insight into the determinants of the cutoff $\bar{\Omega}_A$.

Corollary 4 (Non-Exporting Type-B Sectors). *In the limit $\gamma \rightarrow 0$, the optimal type-A tariff is strictly negative as $\psi_B \rightarrow 0$. More generally, if $0 < \gamma < \Sigma$, the threshold $\bar{\Omega}_A$ is increasing in ψ_B .*

When the export share of type-B sectors is smaller, P_B is less sensitive to exchange-rate movements, provided $\gamma \leq \Sigma$. Appreciation is then less effective at lowering the CPI, so a larger appreciation is needed to offset tariff-induced inflation, making the tariff more likely to backfire.⁴⁰

Corollary 5 (High Export Exposure in Type-A Sectors). *For any $\Omega_A > 0$, the optimal policy is an import subsidy in the limit $\psi_A \rightarrow 1$.*

When a sufficiently large share of type-A output is exported, the contraction in export demand caused by appreciation outweighs the gain from import substitution, making an import subsidy optimal. Conversely, when the export share is small, the export-contraction channel is weak and tariffs are less likely to backfire.

Magnitude of the Optimal Tariff The employment response underlying Theorem 2 also lets us bound the magnitude of the optimal tariff. Let $\tau_A \equiv t_A/(\epsilon p_{mA}^*)$ and $\mathcal{M}_A \equiv \mu_A/w_A$ denote the type-A tariff and labor wedge in ad valorem terms, respectively. Because the currency appreciation channels are contractionary, they reduce the optimal tariff relative to the case with only the direct import-substitution effect. Isolating this direct effect therefore yields an upper bound on the optimal tariff:

$$\frac{\tau_A}{1 + \tau_A} \leq \mathcal{M}_A \frac{(\sigma_A - \Sigma)(1 - \omega_A)}{\sigma_A(1 - \omega_A) + \Sigma\omega_A}. \quad (28)$$

40. Appendix B.2 provides an interpretation of Corollary 4 through the lens of generalized Lerner symmetry.

The bound has a natural benefit-cost interpretation. The numerator reflects how strongly a higher import price shifts expenditure toward domestic goods, while the denominator reflects how strongly it distorts imports. Accordingly, the bound is increasing in the within-sector elasticity σ_A : as σ_A approaches Σ , the import-substitution motive vanishes and the bound converges to zero.

The bound becomes tighter as the expenditure share of type- A sectors becomes small and is exact in the infinitesimal-type- A limit under linear technology. The same bound applies sector by sector even when we relax the as-if two-sector restriction in Definition 3, provided the upper-level aggregator H is CES with elasticity Σ .

Other Wedges Other distortions—such as markup distortions arising from sticky prices and other externalities—enter as additive terms in the Pigouvian optimal tariff formula. The expression in (22) continues to capture the component of the optimal tariff arising from the motive to address labor wedges. The model features introducing these distortions (e.g., price stickiness) may nevertheless change the expression for the employment response to import restrictions entering the labor-wedge component. The currency-pricing extension in Section 3.4 illustrates this point.

3.4 Extensions

We briefly discuss several extensions of the baseline analysis; Appendix B provides a full treatment. The Pigouvian approach is quite general and extends to all the settings considered below, although additional distortions may introduce additional terms into the optimal-policy formula in Theorem 1.

Input-Output linkages Our baseline treatment assumes production takes a horizontal structure, without intermediate inputs. However, it is well known that the presence of intermediate inputs changes the efficacy of tariffs. In Appendix B.4 we show that extending the theoretical model to account for input-output linkages further reduces the case for tariffs. The reason is twofold: first, tariffs on imported inputs translate into higher domestic output prices, reducing demand. Second, when a sector j uses other inputs in its production, the increase in labor demand due to stimulus of sector j is muted, as demand "leaks" to other inputs. Both of these forces apply absent an exchange-rate response, and so are somewhat orthogonal to our main results of the previous section. We

nonetheless include input-output linkages in our quantitative analysis, and find that such linkages play an important role in reducing the effectiveness of tariffs even in a sector-specific recession.

Export Instruments Our baseline assumes that export subsidies are unavailable. When available, they provide a more direct way to support employment in sectors where DNWR binds, since they increase foreign demand for their output without creating as much inflationary pressure. Export subsidies therefore serve as a partial substitute for sector-specific labor subsidies. Theorem B.2 characterizes optimal policy when export subsidies are also available.

Restricted Tariff Instruments Theorem 2 allows tariffs on both sector types in the short run and the long run. If, instead, long-run tariffs are unavailable and fixed at zero, the theorem continues to hold, with $\check{\gamma}$ defined holding long-run tariffs rather than long-run imports fixed. If, furthermore, short-run tariffs are available only in type-A sectors, the sign of the optimal type-A tariff continues to be characterized by a cutoff condition, with a different pass-through term replacing Ξ_B .

Many Sectors We also consider optimal policy with many sectors, retaining the as-if two-period structure of Definition 2 and assuming that $H(\cdot)$, $h_j(\cdot)$, and $G(\cdot)$ are CES and $f_j(\cdot)$ is linear, as is standard in applied work. As above, let $\mathcal{A}$ denote the sectors in which DNWR binds. The sign of the optimal tariff on each sector $j \in \mathcal{A}$ reflects the same tradeoff as in the as-if two-sector case: import substitution stimulates employment *within* that sector, but the induced currency appreciation reduces employment *across* sectors where DNWR binds through its contractionary effect on exports and aggregate demand.

The combined expenditure share of sectors with binding DNWR, $\Omega_{\mathcal{A}}$, continues to be a key determinant of the contractionary effects of appreciation. In the special case with a common ad valorem labor wedge $\mathcal{M}_j \equiv \mu_j/w_j$ across sectors in $\mathcal{A}$, optimal policy converges to import subsidies in all sectors as $\Omega_{\mathcal{A}} \rightarrow 1$. At the opposite limit, as $\Omega_{\mathcal{A}} \rightarrow 0$, it prescribes targeted tariffs on the sectors where DNWR binds, provided $\sigma_j > \Sigma$.

When the labor wedges $\mathcal{M}_j$ differ across sectors, an additional consideration arises: among sectors in $\mathcal{A}$, it is welfare-improving to reallocate labor from lower- to higher-wedge sectors. As a

result, the optimal tariffs need not be negative in every sector even in the limit as $\Omega_{\mathcal{A}} \rightarrow 1$. However, in this limit at least one sector will have negative optimal tariffs. Theorems B.7 and B.9 provide explicit formulas for optimal tariffs with common and heterogeneous labor wedges, respectively.

Currency of Pricing The analysis above assumes complete exchange-rate pass-through, abstracting from currency-pricing frictions emphasized by Devereux and Engel (2003) and Gopinath et al. (2020). One might expect the adverse export-demand effect of appreciation to be smaller under local currency pricing (LCP) of exports—that is, when export prices are sticky in the currency used by foreign buyers—since appreciation passes through only partially to prices abroad. But this reasoning is incomplete: under CPI inflation targeting, export-side LCP weakens the decline in P_B caused by a given appreciation, so the currency must appreciate more strongly to offset the inflationary pressure from the tariff-induced rise in p_{mA} .

We establish the following results under linear technology, using a first-order approximation around the initial symmetric steady state. In the limit $\gamma \rightarrow 0$, the larger appreciation exactly offsets the weaker export response, leaving the employment-import response unchanged. With $\gamma > 0$, exact neutrality no longer holds because real interest parity translates the larger appreciation into a higher domestic real interest rate, further depressing current consumption. Nevertheless, the departure from neutrality remains modest when either $\check{\gamma}$ or the degree of export-side LCP is small.

As discussed in Section 3.2, pricing distortions introduce additional terms into the Pigouvian formula, so the expression in (22) now characterizes the component of optimal policy arising from the motive to address labor wedges. Theorem B.11 extends Theorem 2 by characterizing the sign of this component, with a similar cutoff condition for type-A sectors.

Alternative Inflation Targets Our baseline analysis assumes that the central bank stabilizes the consumption price index, as consumer prices are the stated target of modern central banks. In our model, consumer prices have a precise definition as the ideal price index of the final consumption aggregate. In practice various indices of consumer prices are used to measure inflation. In Appendix B.9, we show that our main qualitative result—that optimal tariffs are negative when sectors with binding DNWR account for a sufficiently large share of the economy—is robust to

alternative measures of consumer-price inflation.⁴¹

Joint Trade and Monetary Policy A natural concern is that the monetary-policy response to trade policy might overturn the results above. We therefore consider a setting where monetary policy responds optimally to trade policy based on a welfare-based dual mandate. We show that the tariff-sign characterization in Theorem 2 continues to hold.

We introduce costly inflation to preclude the uninteresting outcome where the central bank generates enough inflation to ensure that all nominal wages rise and DNWR no longer binds. The welfare formula in equation (21) must now incorporate the social marginal cost of inflation, denoted by $\phi_t^0(\pi_t)$ in period-0 monetary units. The first-order welfare variation becomes

$$d\mathcal{W}_0 = \sum_{t=0}^{\infty} \sum_{j \in \mathcal{J}} \mu_{jt}^0 d\ell_{jt} + \sum_{t=0}^{\infty} \sum_{j \in \mathcal{F}_t} t_{jt}^0 dm_{jt} - \sum_{t=0}^{\infty} \phi_t^0(\pi_t) d\pi_t. \quad (29)$$

Appendix B.10 specifies the inflation cost.

In this setting, we consider a Stackelberg game where the government first sets the path of tariffs, anticipating the central bank's subsequent choice of inflation. We assume both authorities share the same objective of maximizing $\mathcal{W}_0$.⁴²

The setting gives rise to a dual mandate for monetary policy. At the Stackelberg equilibrium, optimal inflation satisfies

$$\phi_s^0(\pi_s^*) = \sum_{t=0}^{\infty} \sum_{j \in \mathcal{J}} \mu_{jt}^0 \left. \frac{\partial \ell_{jt}}{\partial \pi_s} \right|_{dm_{\mathcal{F}}=0}. \quad (30)$$

In words, monetary policy balances the marginal cost of inflation—the left-hand side—against the welfare gains from higher employment in sectors with labor wedges—the right-hand side. Thus, the central bank responds endogenously to trade policy based on this “dual mandate.”

Strikingly, even though we no longer assume strict inflation targeting, Theorem 2 continues to characterize the signs of optimal tariffs, with shares and elasticities evaluated at the Stackelberg equilibrium. Intuitively, since monetary policy is optimized, the central bank's response to trade

41. This result requires the inflation target to retain some weight on import prices as the economy approaches the one-sector limit. In the knife-edge case where the central bank stabilizes domestic producer prices alone, tariffs have no impact on employment in the limit, and the optimal tariff converges to zero.

42. As a result, the outcome coincides with the joint optimum over trade and monetary policy.

policy has no first-order welfare impact. Hence, the Pigouvian formula (22), with the inflation path held fixed at the Stackelberg equilibrium, continues to characterize optimal tariffs.

Anticipated Downturns and Prudential Trade Policy In the general multi-period model, an additional policy motive arises when the planner expects DNWR to bind in sector j at $t + 1$ but not at t : lowering $w_{j,t}$ relaxes the future constraint by reducing $\bar{w}_{j,t+1} \equiv \delta_j w_{j,t}$. This creates a prudential motive for trade policy similar to Schmitt-Grohé and Uribe (2016). During a boom, tariffs—which by Lerner symmetry function as export taxes combined with tighter capital controls—can restrain wage growth and make DNWR less restrictive in a future recession.

4 Quantitative Analysis

We now use the calibrated model to quantify the mechanisms identified in the theory. We contrast an aggregate demand shock, which creates labor wedges across many industries, with an automotive trade shock, which concentrates the distortion in one industry. The two exercises illustrate the theoretical results and assess the magnitudes of the optimal interventions.

4.1 Quantitative Model

Preferences and demand. Period utility is

$$u_t(C_t, L_t) = \frac{C_t^{1-1/\gamma}}{1-1/\gamma} - b \frac{L_t^{1+1/\varphi}}{1+1/\varphi}, \quad (31)$$

where γ is the intertemporal elasticity of substitution, φ the Frisch elasticity, and b the labor-disutility scale. The final-consumption aggregator H_t is Cobb–Douglas, and h_j is CES with elasticity σ_j .

Production and input–output linkages. The domestic production structure follows Caliendo and Parro (2015). Sectoral output is

$$y_{jt} = z_{jt} \ell_{jt}^{\alpha_j^\ell} \prod_i M_{ijt}^{\alpha_{ij}^M}, \quad \alpha_j^\ell + \sum_i \alpha_{ij}^M \leq 1, \quad (32)$$

where fixed sectoral capital is absorbed into z_{jt} and M_{ijt} denotes a composite intermediate input, which is a CES combination of domestic d_{ijt}^{input} and foreign inputs m_{ijt}^{input} , with elasticity of substitution σ . We assume a common elasticity in all sectors, but allow the CES weights to differ by use and, for intermediate purchases, across purchasing industries.

Dynamic labor supply. We propose a novel model of dynamic sectoral choice that extends the Roy–Fréchet model while allowing for saving and nonlog utility. The popular dynamic discrete-choice model of Caliendo et al. (2019) is not tractable with saving, motivating our alternative.⁴³

In particular, we specify the labor-supply frontier as

$$G_t(\ell_t; \ell_{t-1}) = \left[\sum_j \left\{ \frac{\ell_{jt}^{1/(1-\chi)}}{\xi_j \ell_{j,t-1}^{\chi/(1-\chi)}} \right\}^{\vartheta/(\vartheta-1)} \right]^{(\vartheta-1)/\vartheta} \leq L_t, \quad \vartheta > 1, \quad 0 \leq \chi < 1. \quad (33)$$

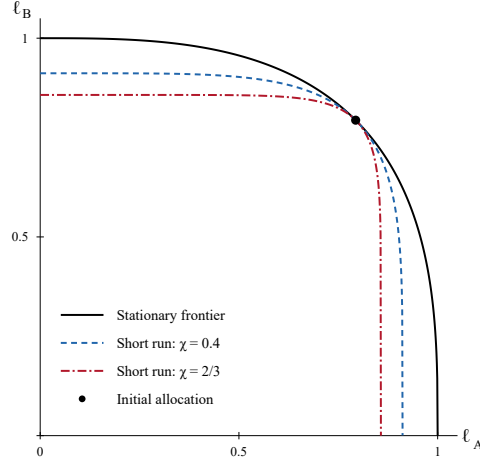
The frontier in a stationary allocation coincides with that of the static Roy–Fréchet model with shape parameter ϑ and sectoral scales ξ_j . However, when $\chi > 0$, the degree of cross-sector substitutability of labor is lower in the short run, with χ governing the curvature of the frontier. Formally, conditional on allocations in the previous period, the elasticity of substitution across sectoral labor inputs is $(1 - \chi)(\vartheta - 1)/[1 + \chi(\vartheta - 1)]$, compared with $\vartheta - 1$ in the long run.

Figure 2 illustrates the frontier for two sectors. The black curve is the stationary frontier. Given the initial allocation indicated by the dot, the blue and red curves show the short-run frontiers for $\chi = 0.4$ and $\chi = 2/3$, respectively. As χ rises, the short-run frontier becomes more curved, indicating a higher degree of mobility friction.

This frontier can be microfounded by combining Roy–Fréchet sectoral choice with gradual reallocation across sectors. Workers draw sector-specific skills independently from Fréchet distributions with shape parameter ϑ . There is a continuum of symmetric jobs within each sector, and

43. In CDP, workers consume current wage income and have log utility, so the sectoral choice problem can be written in log real wages plus additive utility shocks. With saving, even under complete insurance, consumption is not tied to a worker’s current wage, so this log-wage representation and its closed-form aggregation no longer apply without further restrictions. Kleinman et al. (2023) accommodate capital accumulation within the CDP framework by separating hand-to-mouth workers from landlords who save. However, for our welfare analysis, this formulation would introduce an additional wedge: workers and capital owners generally have different intertemporal marginal rates of substitution. Our formulation instead allows us to focus on labor wedges arising from DNWR.

Figure 2: Short-run and stationary labor-supply frontiers



Notes: The illustration sets $\vartheta = 1.5$, $\xi_A = \xi_B = 1$, and $L = 1$. The dot marks the symmetric initial allocation. The black curve is the stationary frontier; the dashed and dash-dotted curves are short-run frontiers for $\chi = 0.4$ and $\chi = 2/3$. Their substitution elasticities are $1/4$ and $1/8$, compared with 0.5 in the stationary allocation.

sectoral effective labor is a Cobb–Douglas aggregate over them. Each period, a randomly selected fraction $1 - \chi$ of jobs becomes available for reallocation, while workers in the remaining jobs must stay in place. Mobile workers choose sectors and jobs according to the present discounted value of expected earnings during their tenure. Aggregating their choices yields equation (33).⁴⁴

Forward-looking labor choice and DNWR. Sluggish adjustment makes current employment valuable for future earning opportunities. Under flexible wages, define the return to employment in sector j , inclusive of this continuation value, by

$$v_{jt} = w_{jt} + \frac{\chi}{1 + i_t} \frac{v_{j,t+1} \ell_{j,t+1}}{\ell_{jt}}. \quad (34)$$

The household equates this return to the marginal cost of labor supply:

$$v_{jt} = P_t b C_t^{1/\gamma} L_t^{1/\varphi} \frac{\partial G_t}{\partial \ell_{jt}}. \quad (35)$$

When DNWR binds, the labor wedge μ_{jt} enters the corresponding condition in equation (20); the continuation value then depends on future wedge-adjusted returns.

44. Appendix C.1 provides the full construction.

4.2 Data

We use the 2024 BEA Input–Output Accounts to construct expenditure, trade, and production shares for 66 U.S. industries. The make and use tables and the corresponding import matrix provide intermediate purchases, final expenditure, imports, exports, value added, and labor compensation. The import matrix allows us to construct separate import shares for final use and for intermediate purchases by each industry.⁴⁵ Appendix D.1 describes the data construction.

4.3 Calibration

We use annual periods and set the household discount factor to $\beta = 0.98$, the IES to $\gamma = 0.5$, and the Frisch elasticity of labor supply to $\varphi = 0.5$, all standard values in the literature. The discount factor implies a stationary real interest rate of about 2 percent. We set the elasticity of substitution between domestic and imported goods and the elasticity of foreign demand for U.S. exports to $\sigma_j = \sigma_j^* = 3$, implying trade elasticities of 2.⁴⁶ For cross-sector labor allocation, we set $\vartheta = 1.5$ following Galle et al. (2023), implying a long-run substitution elasticity of 0.5; we set the persistence parameter to $\chi = 0.4$, reducing the short-run elasticity to 0.25. Finally, we set the wage-floor parameter $\delta_j = 0.99$ in every industry, drawing on the calibrated value for U.S. manufacturing in Rodriguez-Clare et al. (2025). Table 1 summarizes the benchmark parameters.

We solve counterfactual equilibria in exact hats relative to a stationary 2024 allocation with zero net tariffs and no labor wedges. BEA expenditure, input–output, and factor-payment shares determine the weights in preferences and production, while BEA import and export values pin down the world import prices and export-demand shifters. Appendix D.2 provides details of the

45. BEA constructs the import matrix by applying a common import-to-domestic-supply ratio to all uses of each detailed commodity. After aggregation from commodities to industries, import shares differ across uses because final users and producing industries purchase different mixtures of detailed commodities.

46. Fajgelbaum et al. (2020) (hereafter FGKK), using the 2018 U.S. tariff hikes, estimate one-month elasticities of 1.19 between domestic and imported goods and 2.53 across foreign varieties within HS-10 products. Our higher value of σ_j reflects adjustment beyond the one-month horizon and also gives tariffs greater scope to shift expenditure toward domestic goods. For the elasticity across foreign varieties, Boehm et al. (2023) estimate roughly 1 at a five-year horizon and 2 at a ten-year horizon, while Teti (2024), using revised tariff data, obtains a ten-year estimate of 0.8. These longer-horizon estimates are all below FGKK’s one-month estimate of 2.53 and the preferred tariff-based estimate of about 5 in the older gravity literature surveyed by Head and Mayer (2014).

Table 1: Benchmark parameters

Parameter	Symbol	Value
Discount factor	β	0.98
Intertemporal elasticity of substitution	γ	0.5
Frisch elasticity	φ	0.5
Domestic–import substitution elasticity	σ_j	3
Export-demand elasticity	σ_j^*	3
Roy–Fréchet shape parameter	ϑ	1.5
Labor-allocation persistence	χ	0.4
Wage-floor persistence	δ_j	0.99

calibration.

As for policy, we assume tariffs can be set in traded merchandise sectors.⁴⁷ In both exercises, monetary policy holds CPI inflation at zero. This makes the case for tariffs favorable: positive inflation would relax DNWR and generally reduce the magnitude of optimal tariffs. We examine positive inflation targets in the appendix.

4.4 Exercise I: Aggregate Demand Shock

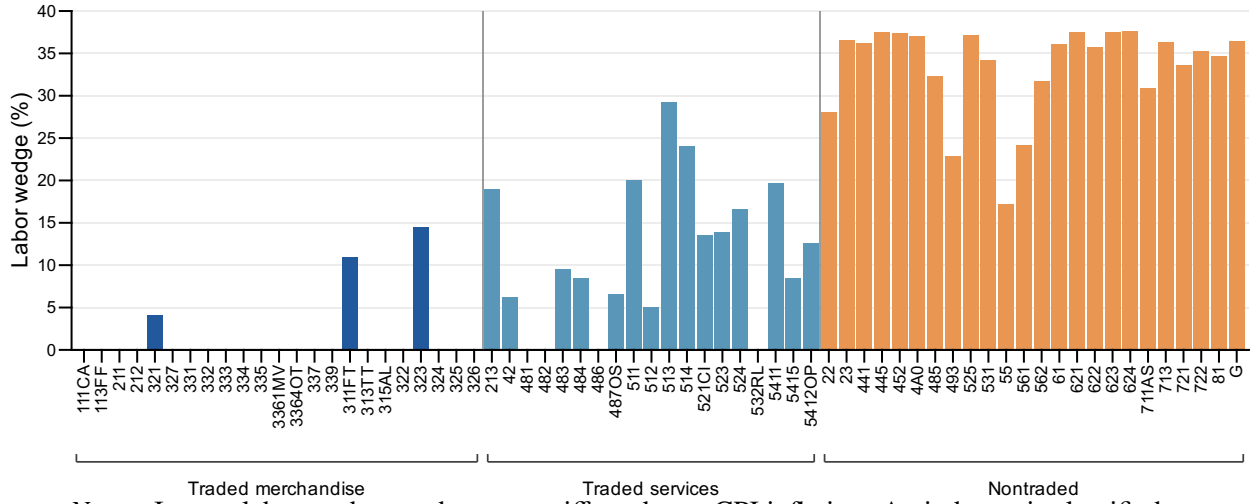
In the first exercise, we revisit Keynes’s rationale for protection during a demand-driven recession. Keynes advocated tariffs during the Great Depression under the gold standard, but our theory predicts import subsidies as the optimal policy in the flexible-exchange-rate setting. We use the quantitative model to assess how large these subsidies are and how much they alleviate DNWR-induced employment losses.

Shock. We consider a transitory increase in the household discount factor. The shock induces the household to shift consumption toward the future, reducing aggregate demand in the short run. Specifically, we assume that the discount factor rises by four log points on impact and returns toward its steady-state value with a four-year half-life:

$$\log(\beta_t) - \log(\beta) = (0.04) 2^{-t/4}, \quad t \geq 0. \quad (36)$$

47. We allow tariffs only in sectors where the combined baseline import share across final and intermediate uses is at least one percent. This cutoff avoids numerical instability in industries with negligible imports.

Figure 3: Labor wedges by industry type and export exposure



Notes: Impact labor wedges under zero tariffs and zero CPI inflation. An industry is classified as traded if its baseline exports are at least 1 percent of gross output. See Table 5 for industry names.

The shock arrives unexpectedly, after which agents foresee its path. We choose a deliberately large shock to generate substantial labor wedges and make the strongest quantitative case for trade intervention.

Labor wedges. Figure 3 shows the labor wedges at impact. The shock produces substantial labor wedges, concentrated in services and especially in nontraded industries. Specifically, industries exporting less than 1 percent of gross output have an average wedge of 33.5 percent, compared with 1.2 percent among industries exporting at least 10 percent. Industries with high export intensity experience less downward pressure on their wages because foreign demand continues to support employment. To give a sense of aggregate magnitudes, the shock reduces the aggregate labor L_t by 2.4 percent at impact. Under flexible wages, L_t would instead rise by 3.0 percent; greater patience induces households to work more and increase saving. Appendix Figure D.1 reports the corresponding sectoral employment shortfalls.

Optimal tariffs and the exchange rate. The blue bars in panel (a) of Figure 4 show the optimal tariff rates at impact. As predicted by the theory, the optimal policy subsidizes imports across sectors, with subsidy rates ranging from 4.3 to 9.3 percent.

The theory predicts that import subsidies boost employment in sectors with binding DNWR

by inducing currency depreciation. Panel (b) of Figure 4 plots the policy-induced exchange-rate response alongside the import-weighted average tariff. At impact, the average tariff is approximately -7 percent, and the real exchange rate depreciates by 2 percent relative to the equilibrium without intervention.

To further highlight the role of the exchange-rate channel, we evaluate the optimal tariff formula (22) while holding the exchange rate fixed ($d\epsilon = 0$) rather than CPI inflation fixed ($d\pi = 0$).⁴⁸ The orange bars show the resulting formula-implied tariffs. With the exchange-rate response shut down, the formula implies positive tariffs ranging from 2.4 to 5.3 percent across industries. The sign reversal confirms that tariff-induced currency appreciation is the key channel making tariffs unsuitable for addressing aggregate downturns.⁴⁹

Welfare. Optimal trade policy recovers only 21 percent of the welfare loss from wage rigidity relative to the first-best allocation. The gain is equivalent to a one-time transfer of 0.07 percent of baseline GDP. Trade policy is far from a close substitute for first-best labor subsidies.

4.5 Exercise II: Automotive Trade Shock

In the second exercise, we consider a sector-specific shock, for which the exchange-rate response to the optimal tariff is expected to be small. Motivated by the rapid expansion of Chinese electric-vehicle production and the resulting increase in competition in global automobile markets, we study a terms-of-trade shock that lowers the world prices of automobiles. The exercise highlights the role of input–output linkages in shaping the optimal tariff.

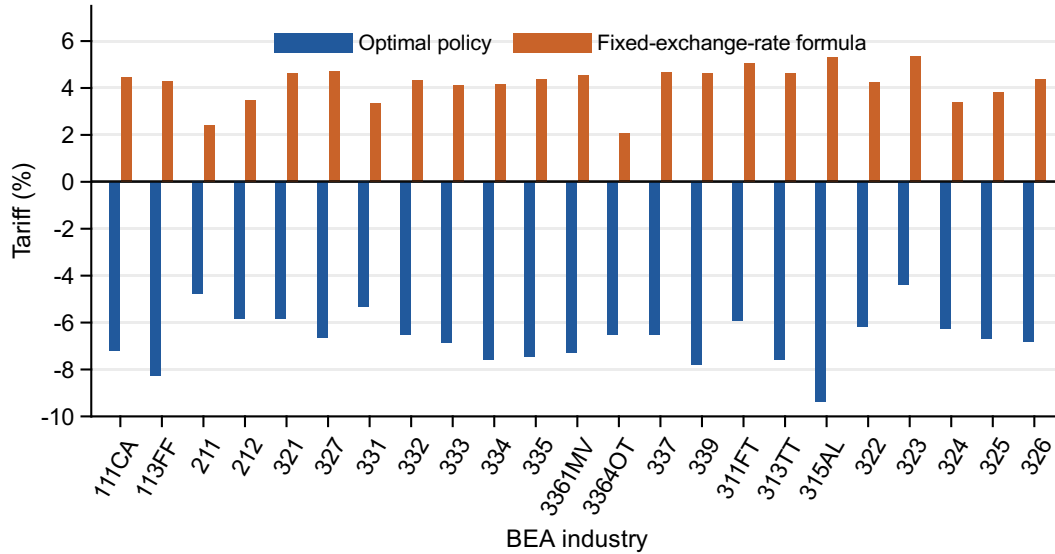
Shock. We model the shock as an immediate and permanent 20 percent decline in the world prices of automotive imports and exports at any given trade allocation. Formally, for the automobile sector, we set $p_{mjt}^* = 0.8p_{mj,-1}^*$ and $\rho_{xjt}^* = 0.8\rho_{xj,-1}^*$ for all $t \geq 0$. Thus, the shock shifts both the import-supply and export-demand schedules proportionally downward.

48. We evaluate the shares and elasticities at the same optimal allocation. Appendix D.3 explains the construction.

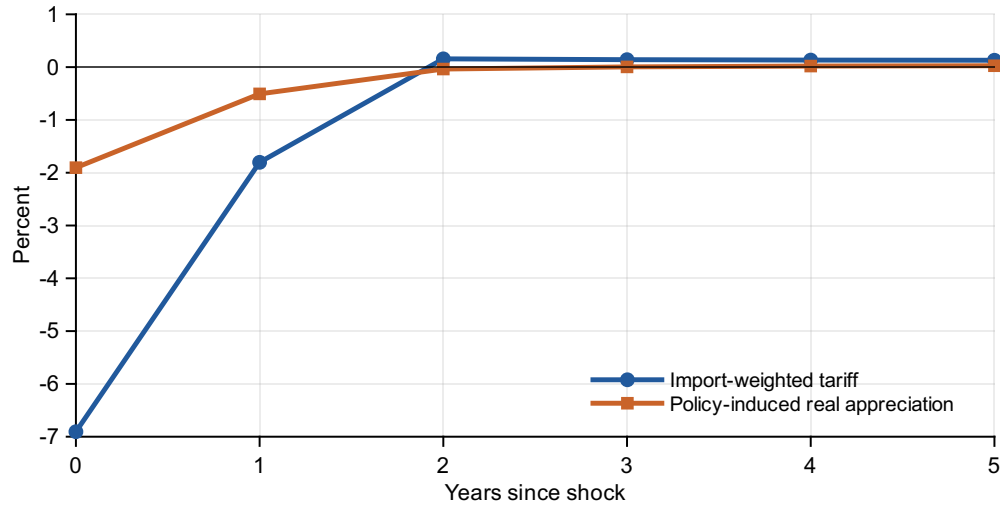
49. Unlike the comparative statics considered in the theoretical analysis, this exercise allows CPI inflation to respond. Trade policy can therefore move the real exchange rate ϵ_t/P_t even when the nominal exchange rate is fixed, inducing intertemporal substitution in consumption through real interest parity (16). Of the two general-equilibrium channels discussed in Section 3.2, the aggregate demand channel thus remains operative.

Figure 4: Optimal import subsidies and the exchange-rate channel

(a) Optimal import subsidies and the fixed-exchange-rate diagnostic



(b) Tariffs and the real exchange rate after the aggregate demand shock



Notes: Panel (a) shows impact tariffs under the optimal policy and the local tariff formula with the exchange-rate response suppressed. Panel (b) shows the import-weighted average tariff and the policy-induced change in the real exchange rate relative to the no-tariff equilibrium. Negative values denote import subsidies and depreciation.

Employment and labor wedges. The first two columns of Table 2 summarize the impact of the shock. At impact, employment in the automobile sector falls by 17.9 percent under DNWR, compared with 3.8 percent in the first-best allocation. Automotive employment is therefore 14.7 percent below its first-best level without intervention, and the associated labor wedge is 70.9 percent. Labor wedges are zero in every other sector except primary metals (BEA code 331), where the

Table 2: Impact automotive outcomes

	First best	No tariff	Optimal tariffs
Automotive employment change (percent)	−3.8	−17.9	−15.6
Automotive employment gap relative to first best (percent)	0	−14.7	−12.3
Automotive labor wedge (percent)	0	70.9	61.3
Welfare loss (percent of baseline automotive value added)	0	5.6	4.9

Notes: Changes are relative to the stationary baseline. The employment gap is measured relative to the first-best allocation; negative values denote shortfalls. The welfare loss is measured relative to the first-best allocation as an equivalent one-time transfer and scaled by baseline automobile-sector value added.

impact wedge is 3.6 percent.

Optimal tariffs. Panel (a) of Figure 5 presents the optimal tariffs at impact, while panel (b) traces the optimal automotive tariff over time alongside the policy-induced real appreciation. The optimal automotive tariff is only 3.8 percent at impact, while all other sectors receive import subsidies below 0.5 percent. The automotive tariff then declines steadily over time. As predicted, the real appreciation is negligible, so the small tariff cannot be attributed to the exchange-rate channel.

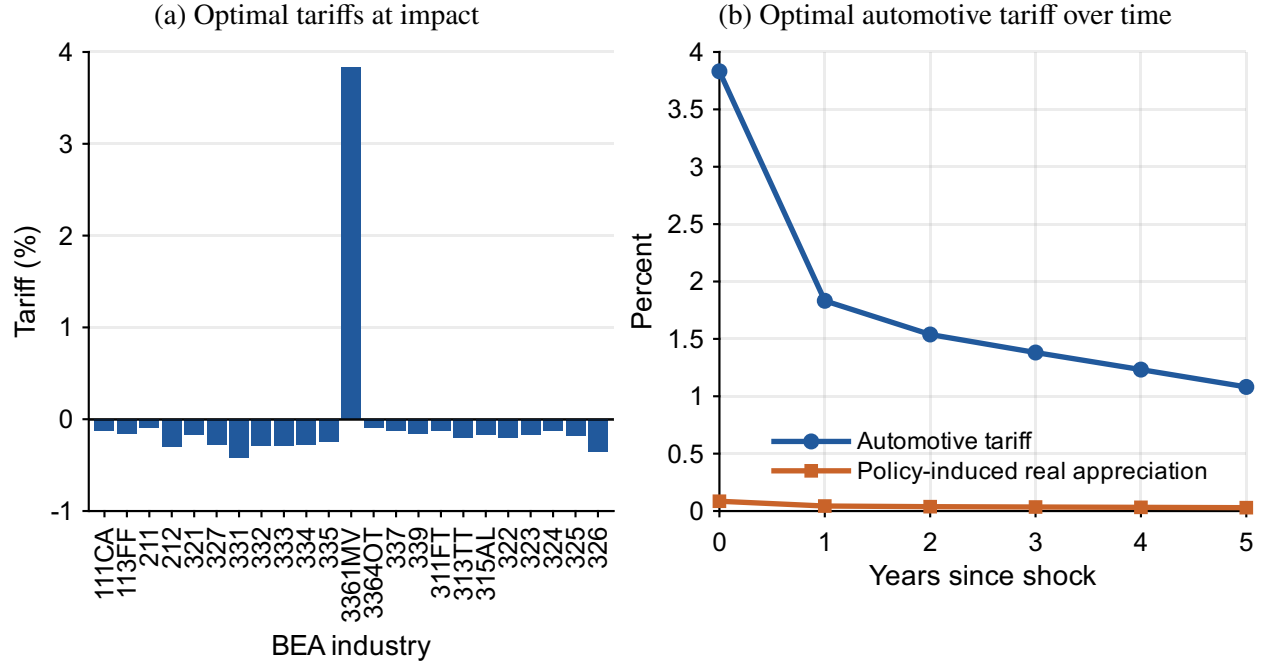
The last column of Table 2 shows the implications for employment and welfare. The tariff raises automotive employment at impact by only 2.3 percentage points, leaving a 61.3 percent labor wedge and employment 12.3 percent below the first-best benchmark. The welfare loss falls from 5.6 to 4.9 percent of baseline automobile-sector value added, so the tariff recovers only about 12 percent of the deadweight loss from DNWR.

Viewed through the Pigouvian formula (22), the small optimal tariff despite the large labor wedge indicates that using tariffs to restrict imports is ineffective at expanding automotive employment. We next examine how input–output linkages make even a targeted tariff a blunt instrument.

Final-good versus input tariffs. Appendix B.4 identifies two ways input–output linkages weaken the effectiveness of a tariff intervention. First, input tariffs raise domestic production costs. Second, spending redirected toward domestic production partly leaks to upstream sectors and imported inputs instead of labor in the targeted sector. We perform two exercises to gauge the quantitative importance of these channels in the calibrated economy.

To explore the first channel, we allow separate tariffs on final and intermediate purchases. If

Figure 5: Optimal tariffs and real appreciation after the automotive shock



Notes: Panel (a) shows impact optimal tariffs across traded-merchandise industries, whose BEA codes and names are listed in Table 5. Panel (b) compares the automotive tariff with the policy-induced real appreciation relative to the no-tariff allocation over five years.

the input-cost channel were quantitatively important, the optimal input tariff would be much lower than the final-good tariff and could become a subsidy.

Table 3 shows the implied optimal automotive tariffs. The first row reports the benchmark with a common tariff, while the second and third rows report the optimal final-good and input tariffs when the two instruments can differ. The optimal input tariff remains positive at 2.87 percent, though below the 4.16 percent final-good tariff; the final-good tariff is itself only slightly above the common tariff. Together, these results suggest that the input-cost channel plays a small role in explaining the low optimal tariff.

Demand leakages. We next isolate the two demand leakages in the import-substitution channel: spending on imported inputs and spending on inputs from upstream sectors.

To isolate the direct import-substitution channel from the general-equilibrium channels, we

Table 3: Optimal automotive tariffs with common and use-specific instruments (percent)

Policy instrument	Impact tariff
Common tariff	3.83
Final-good tariff	4.16
Input tariff	2.87

Notes: The common tariff is from the benchmark policy. The final-good and input tariffs are jointly optimized when the two uses can be taxed separately. All entries are for the automotive industry at impact.

consider a partial-equilibrium counterpart of the Pigouvian formula:

$$\tau_{it}^{\text{PE}} = \frac{1}{\epsilon_t p_{mit}^*} \sum_{j \in \mathcal{J}} \mu_{jt} \left(- \frac{\partial \ell_{jt}}{\partial m_{it}} \Big|_{dC_t=d\epsilon_t=d\mathbf{w}_t=0} \right). \quad (37)$$

The benchmark abstracts from changes in the exchange rate, aggregate consumption, and sectoral wages, but retains the price and sales adjustments transmitted through the production network. Appendix D.3 details the derivative calculation and the four Leontief specifications below.

In our exercise, the labor wedge is concentrated in the automobile sector. Thus, τ^{PE} depends primarily on how automobile employment responds to a restriction on automobile imports. The Leontief inverse determines how the additional demand generated by import substitution propagates across domestic producers. Its automobile-to-automobile entry generalizes the own-sector propagation term in equation (B.24).

To assess the quantitative importance of the two leakages, we evaluate τ^{PE} after modifying the Leontief inverse to shut down each channel in turn. First, we eliminate leakage through imported inputs by setting $\omega_{ij}^{\text{int}} = 0$ in the Leontief inverse. Second, we eliminate leakage to upstream sectors by replacing the input–output matrix in the Leontief inverse with a diagonal matrix. This directs each industry’s domestic intermediate spending back to itself without changing the total represented in the Leontief inverse. Finally, we eliminate both leakages simultaneously.⁵⁰

In the baseline calibration, imports account for 41.0 percent of final automotive use ($\omega_j^{\text{fin}} = 0.410$) and 32.0 percent of the automotive inputs used by automobile producers ($\omega_{jj}^{\text{int}} = 0.320$).

50. We evaluate each local derivative at the same optimal-tariff equilibrium, changing only the Leontief inverse. Expenditure shares and labor wedges are therefore common to all four cases; we do not solve new equilibria under alternative input–output structures.

Table 4: Partial-equilibrium automotive tariff under alternative input–output networks (percent)

	Baseline ω^{int}	Zero ω^{int}
Full IO	4.2	4.9
Diagonal IO	7.7	13.4

Notes: Entries report τ^{PE} for the automobile sector, in percent, evaluated at the same jointly optimal allocation. The four cases modify only the Leontief inverse. “Diagonal IO” assigns each sector’s intermediate-input purchases to its own sector; “Zero ω^{int} ” sets imported-input shares to zero. All other coefficients retain their evaluated values. The full-equilibrium optimal automobile tariff is 3.83%.

Automotive inputs account for 28.6 percent of the sector’s production costs ($\alpha_{jj}^M = 0.286$).

Table 4 shows the results. The baseline τ^{PE} is 4.2 percent, close to the full-equilibrium optimal tariff of 3.83 percent. Eliminating import leakage raises τ^{PE} to 4.9 percent, while replacing the full network with a diagonal network raises it to 7.7 percent. Eliminating both leakages simultaneously raises it to 13.4 percent, indicating that the two channels reinforce each other. Together with our preceding comparison of input and final-good tariffs, these results suggest that demand leakages are an important reason why the optimal tariff is small.

5 Conclusion

We revisited the Keynesian rationale for tariffs: when DNWR creates inefficient employment losses, tariffs can serve as second-best instruments by redirecting labor demand toward sectors with high labor wedges. Our analytical results show, however, that tariffs can be weak or counterproductive instruments for addressing these losses: the exchange-rate response can cause broad interventions to backfire, while input–output linkages weaken the effects of targeted tariffs on sectoral labor demand. In our calibrated U.S. economy, optimal interventions and their welfare gains remain modest even when labor wedges are sizable.

Our analysis abstracts from distributional objectives. Tariffs might appear more attractive if policymakers place additional value on employment in sectors where DNWR causes job losses, thereby redistributing toward affected workers. Yet the mechanisms identified here remain relevant: tariffs may backfire and reduce employment in the sectors they are intended to support, and even

when they do not, they may not be very effective at creating jobs in those sectors. Distributional objectives may strengthen the case for supporting particular sectors, but they do not resolve the question of whether tariffs are effective instruments for providing that support.

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Table 5: BEA codes for all model industries, by industry type

BEA code	Industry
<i>Traded merchandise</i>	
111CA	Farms
113FF	Forestry, fishing, and related activities
211	Oil and gas extraction
212	Mining, except oil and gas
321	Wood products
327	Nonmetallic mineral products
331	Primary metals
332	Fabricated metal products
333	Machinery
334	Computer and electronic products
335	Electrical equipment, appliances, and components
3361MV	Motor vehicles, bodies and trailers, and parts
3364OT	Other transportation equipment
337	Furniture and related products
339	Miscellaneous manufacturing
311FT	Food and beverage and tobacco products
313TT	Textile mills and textile product mills
315AL	Apparel and leather and allied products
322	Paper products
323	Printing and related support activities
324	Petroleum and coal products
325	Chemical products
326	Plastics and rubber products
<i>Traded services</i>	
213	Support activities for mining
42	Wholesale trade
481	Air transportation
482	Rail transportation
483	Water transportation
484	Truck transportation
486	Pipeline transportation
487OS	Other transportation and support activities
511	Publishing industries, except internet (includes software)
512	Motion picture and sound recording industries
513	Broadcasting and telecommunications
514	Data processing, internet publishing, and other information services
521CI	Federal Reserve banks, credit intermediation, and related activities
523	Securities, commodity contracts, and investments
524	Insurance carriers and related activities
532RL	Rental and leasing services and lessors of intangible assets
5411	Legal services
5415	Computer systems design and related services
5412OP	Miscellaneous professional, scientific, and technical services
<i>Non-traded</i>	
22	Utilities
23	Construction
441	Motor vehicle and parts dealers
445	Food and beverage stores
452	General merchandise stores
4A0	Other retail
485	Transit and ground passenger transportation
493	Warehousing and storage

BEA code	Industry
525	Funds, trusts, and other financial vehicles
531	Real Estate
55	Management of companies and enterprises
561	Administrative and support services
562	Waste management and remediation services
61	Educational services
621	Ambulatory health care services
622	Hospitals
623	Nursing and residential care facilities
624	Social assistance
711AS	Performing arts, spectator sports, museums, and related activities
713	Amusements, gambling, and recreation industries
721	Accommodation
722	Food services and drinking places
81	Other services, except government
G	Government